Andhra University College of Engineering

Civil Engineering | Course Code: CE301

Structural Analysis - Laboratory Manual & Practical Guide

Syllabus Focus: CE301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Energy Theorems & Deflections

" Syllabus Scope: Castigliano first and second theorems, Unit load method, Strain energy in bending and t

Unit 2: Slope Deflection Method

" Syllabus Scope: Derivation of slope deflection equations, Analysis of continuous beams with support set

CE301: Structural Analysis - Advanced Topics & Assessment

Academic Track: Civil Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Moment Distribution Method

" Stiffness and carry-over factors, Distribution factors, Analysis of continuous beams and portal frames witho

Unit 4: Moving Loads & Influence Line Diagrams

" Influence lines for reaction, shear force, and bending moment in simply supported girders, Muller-Breslau pr

Unit 5: Arches & Cable Structures

" Three-hinged and two-hinged circular and parabolic arches, Normal thrust, Radial shear, Bending moment diagr

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Verification of Maxwell Reciprocal Theorem on a Deflected Cantilever Beam

Lab 2: Experimental Determination of Horizontal Thrust in a Three-Hinged Parabolic Arch

Lab 3: Deflection Measurement of a Pin-Jointed Plane Truss under Concentrated Point Loads

Lab 4: Moment Distribution Analysis on a Two-Span Continuous Steel Beam Test Frame

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* C.S. Reddy - Basic Structural Analysis Tata McGraw-Hill 3rd Ed
* R.C. Hibbeler - Structural Analysis Pearson 10th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students